

Jaime Alonso

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EDUCATION

Georgia Institute of Technology

May 2028

B.S. Computer Science — Threads: Systems Architecture & Information Internetworks | **GPA: 4.0/4.0** | Faculty Honors

Honors: Nova 111 Spain 2026 — Ranked Top 10 Nationally in Computer Science (111 selected nationwide)

Coursework: Data Structures & Algorithms, Systems & Networks, Computer Organization, Linear Algebra; upcoming: Design of Operating Systems, Computer Networking

TECHNICAL SKILLS

Languages: C++, C, Python, Kotlin, Java, JavaScript, HTML/CSS

Systems: ARM big.LITTLE, embedded firmware, Linux (syscalls, scheduling, thread affinity), POSIX threads, multithreading/concurrency, performance optimization, on-device inference (GGML, quantized models), Docker, gRPC

Machine Learning & Data: PyTorch, scikit-learn, XGBoost, JAX, Pandas, NumPy, SciPy, Hugging Face, Streamlit

Tools: Git, GitHub Actions (CI/CD), automated testing, CircuitSim, MySQL, Django, Google Cloud Platform

EXPERIENCE

Systems & Infrastructure Engineering Intern | Plix AI

San Mateo, CA | May 2026 — Aug. 2026

AI body cameras for frontline security and field-operations teams; backed by Sequoia Capital and a16z

- Cut projected infrastructure costs ~\$500K/yr by accelerating on-device speech recognition (ASR) inference ~10x (10–37s → 2.5s per window) with an ARM-optimized quantized model, eliminating per-camera cloud transcription
- Shipped real-time incident detection to production embedded firmware on 2,500+ deployed devices, cutting worst-case detection latency 70s → 13s via sliding-window inference scheduling
- Reduced indoor location drift 81% with GNSS/INS sensor fusion (Kalman filter, chi-squared outlier gating, pedestrian dead reckoning), tagging measured vs. inferred positions for provenance in evidentiary records
- Cut audio/video sync drift 37.5s → 0.05s with 0.001% frame loss by re-engineering camera exposure control, validated with automated telemetry-replay tests and multi-hour night field trials

Undergraduate Researcher | LLAMAs Lab, Georgia Institute of Technology

Atlanta, GA | Jan. 2026 — Present

- Reduced Whisper inference latency 88% across wearable targets by tuning CPU core affinities, thread counts, and context window sizes through a C++/Kotlin harness that uses raw Linux syscalls (sched_setaffinity) to bypass Android NDK limits
- Engineered a concurrent Android H.265 video + audio transcription pipeline (Camera2/MediaCodec/MediaMuxer) with thread affinity pinned per workload and CountdownLatch synchronization
- Benchmarked Whisper latency, battery drain, and transcription accuracy across pipeline configurations on ARM big.LITTLE

VIP Research Assistant | Thad Starner's Lab, Georgia Institute of Technology

Atlanta, GA | Aug. 2025 — Jan. 2026

- Identified behavioral risk patterns in construction-safety simulation data (NumPy, Pandas, SciPy) to guide new workers on avoiding and responding to hazards, contributing to a 32% decrease in projected safety incidents

Undergraduate Research Assistant | Gleason Lab, Georgia Institute of Technology

Atlanta, GA | Jan. 2025 — Sep. 2025

- Built a classification pipeline (XGBoost, Random Forest, Logistic Regression) with feature engineering and k-fold cross-validation, predicting cephalopelvic disproportion at 97.25% precision on 6,000+ patients to give clinicians early warning before labor

PROJECTS

Pipelined Processor | CircuitSim, Python

Jan. 2026 — May 2026

- Designed and built a 5-stage pipelined processor (IF/ID/EX/MEM/WB) from scratch in CircuitSim with a dual-ported register file, achieving 4.5–4.8x speedup over the baseline single-cycle datapath
- Implemented full hazard handling — data forwarding for RAW hazards, load-use hazard detection with automatic stalling, and branch-mispredict squashing (predict-not-taken) — as an optional extension beyond the course baseline

Carpet Game Tournament — RatonPerez Agent | Python, NumPy, JAX

Mar. 2026 — May 2026

- Placed top 20 in the live bytefight.org tournament with an agent combining an HMM with Bayesian belief updates over a 64-cell rat location distribution and a negamax engine (PVS, alpha-beta pruning, Zobrist hashing, LMR, tuned move ordering)
- Iterated 6+ agent versions via 50-game batch A/B testing; LMR alone raised average search depth from 8 to 12 ply in real time

Personalized News Digest System | Python, Groq, Llama 3.3, GitHub Actions

Nov. 2025 — Dec. 2025

- Built a serverless news digest pipeline that pulls articles from 10 newspapers via RSS feeds and news APIs, ranks them by relevance with Llama 3.3 (via Groq), and compiles a personalized digest; runs as a scheduled GitHub Actions workflow

LEADERSHIP & COMMUNITY

- **Ramblin' Rocket Club** — Deputy Lead, Software & Simulations: leading simulation architecture for high-powered rocketry
- **SGA** — CS Rep & IT Committee | **GT SHPE** — Volunteer | **Languages:** English, Spanish (Native), French (Fluent)
- **Rocket League** — Ranked top 0.1% globally (Grand Champion)